

Code Layout, Readability & Reusability

Project Development Phase • AI/ML and GenAI Track

Project structure

Path	Purpose
app/main.py	Creates the FastAPI app, mounts static files, registers routes and exception handlers
app/routes.py	HTTP endpoints and the admin authentication helper
app/schemas.py	Pydantic input models with validation rules
app/database.py	SQLAlchemy engine, User model, CRUD helpers with a session context manager
app/gemini_client.py	One reusable generate_text() with retry/backoff and GeminiError
app/gemini_generator.py	Workout plan generation and plan-update prompts
app/gemini_flash_generator.py	Nutrition tip generation
templates/	index, result, all_users and error pages
static/	style.css and app.js
Dockerfile, requirements.txt, README.md	Deployment and setup

Readability practices

- Separation of concerns: routing, validation, AI calls and data access live in separate modules.
- Type hints and small single-purpose functions such as upsert_user, save_updated_plan and get_user.
- Clear names and short docstrings on key functions.
- Configuration through environment variables with sensible defaults.

Reusability practices

- One shared generate_text() wraps every Gemini call, so retry, timeout and error handling are written once.
- A shared _render_result() helper builds the result page for both generation and feedback.
- A shared FORMAT_RULES string keeps the output style consistent across prompts.
- The database URL can be changed to move from SQLite to another database.